

CYBER SECURITY– TASK 4

MULTI-FACTOR AUTHENTICATION

1. INTRODUCTION

Multi-Factor Authentication (MFA) is a security mechanism that requires a user to provide two or more different authentication factors to verify their identity before accessing an account or system.

Traditional authentication often relies only on a username and password. However, passwords can be stolen through phishing, malware, data breaches, or other attacks. MFA provides an additional layer of security by requiring another factor besides the password.

For example, when logging into an account, a user may enter a password and then approve a login request on their mobile device. Even if an attacker obtains the password, access may still be blocked because the attacker does not possess the second authentication factor.

2. OBJECTIVE

The main objective of this task is to understand Multi-Factor Authentication and the different authentication factors used to verify a user's identity.

Specific Objectives

1. Understand the concept of Multi-Factor Authentication.
 2. Identify different types of authentication factors.
 3. Understand the difference between authentication factors.
 4. Compare different MFA methods.
 5. Understand how MFA improves account security.
 6. Identify the limitations of different authentication methods.
 7. Understand whether an OTP alone qualifies as MFA.
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3. WHAT IS AUTHENTICATION?

Authentication is the process of verifying the identity of a user, device, or system before granting access to a resource.

For example, when a user enters a username and password to access an email account, the system checks whether the provided credentials belong to the user.

Authentication is different from authorization.

Authentication: Verifies who the user is.

Authorization: Determines what the authenticated user is allowed to access or perform.

4. WHAT IS MULTI-FACTOR AUTHENTICATION?

Multi-Factor Authentication is an authentication method that uses **two or more independent authentication factors** from different categories.

The three major authentication factor categories are:

1. **Something You Know**
2. **Something You Have**
3. **Something You Are**

For MFA to be effective, the factors should be independent. Simply using two pieces of information from the same category does not necessarily provide true multi-factor authentication.

5. AUTHENTICATION FACTORS

5.1 Something You Know

This factor is based on information that the user knows.

Examples:

- Password
- PIN
- Passphrase
- Security question

Example:

A user enters their password while logging into an account.

Advantages:

- Easy to implement
- Familiar to users
- Relatively inexpensive

Disadvantages:

- Can be guessed

- Can be stolen through phishing
 - Can be exposed through data breaches
 - Users may reuse passwords
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6. SOMETHING YOU HAVE

This factor is based on something the user possesses.

Examples:

- Mobile phone
- Hardware security key
- Authentication device
- Smart card
- Authenticator application

Example:

A user enters their password and then enters a verification code generated by an authenticator application.

Advantages:

- Provides an additional security layer
- Can protect accounts even if the password is compromised
- Hardware security keys can provide strong phishing-resistant authentication

Disadvantages:

- Device can be lost or stolen
 - Some methods may depend on network or mobile availability
 - Users may have difficulty accessing accounts if the device is unavailable
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7. SOMETHING YOU ARE

This factor is based on a user's biometric characteristics.

Examples:

- Fingerprint
- Facial recognition

- Iris recognition
- Other biometric characteristics

Example:

A user enters a password and then verifies their identity using a fingerprint.

Advantages:

- Convenient
- Difficult for another person to reproduce exactly
- User does not need to remember the biometric characteristic

Disadvantages:

- Privacy concerns
 - Biometric information cannot simply be changed like a password
 - Sensors may sometimes fail to recognize the user
 - Biometric systems require appropriate security and privacy controls
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8. OTHER AUTHENTICATION-RELATED FACTORS

Some authentication systems also consider additional signals when determining whether a login should be allowed.

Location

The system may consider where the login originates.

Device

The system may recognize whether the user is accessing the account from a known device.

Behaviour

The system may analyze unusual login patterns or activity.

These signals can improve security, but they are not automatically equivalent to one of the three traditional authentication factor categories.

9. COMMON MFA METHODS

9.1 SMS OTP

A One-Time Password is sent to the user's registered mobile number through SMS.

Process:

1. User enters username and password.
2. System sends an OTP through SMS.
3. User enters the OTP.
4. System verifies the OTP.
5. Access is granted.

Advantages:

- Easy to understand
- Widely available
- Does not require a dedicated authentication application

Limitations:

SMS-based authentication can be exposed to risks such as phishing, SIM-related attacks, and interception.

10. AUTHENTICATOR APPLICATION

Authenticator applications generate time-based or event-based verification codes.

Examples include applications that generate temporary codes for account login.

Process:

1. User enters username and password.
2. The system requests a verification code.
3. User opens the authenticator application.
4. User enters the current code.
5. The system verifies the code.
6. Access is granted.

Advantages:

- More secure than relying on SMS alone in many situations
- Can work without cellular SMS delivery
- Codes change regularly

Limitations:

- The device can be lost.

- Users need access to the authenticator application.
 - Codes can still potentially be entered into phishing websites.
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11. PUSH NOTIFICATION

In push-based MFA, the authentication system sends a login approval request to a registered device.

Process:

1. User enters their username and password.
2. A notification appears on the registered device.
3. User reviews the login request.
4. User approves or denies the request.
5. Access is granted if the request is approved.

Advantages:

- Convenient
- Easy for users
- Faster than manually entering codes

Limitation:

Users may accidentally approve fraudulent login requests if attackers repeatedly send authentication requests. This is sometimes called **MFA fatigue or push bombing**.

12. HARDWARE SECURITY KEY

A hardware security key is a physical device used to authenticate a user.

The user connects or taps the security key when logging in.

Advantages:

- Provides strong authentication
- Can provide strong protection against phishing
- Does not depend on SMS delivery
- Difficult for a remote attacker to use without possessing the physical key

Disadvantages:

- Can be lost or damaged

- Requires compatible systems
 - Organizations may need backup authentication methods
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13. BIOMETRIC AUTHENTICATION

Biometric authentication verifies a user using physical characteristics such as a fingerprint or face.

Example:

A user enters their password and confirms their identity using fingerprint authentication.

Advantages:

- Convenient
- Difficult to share
- No password needs to be memorized for the biometric factor

Disadvantages:

- Privacy considerations
 - False acceptance or rejection can occur
 - Biometric information requires careful protection
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14. MFA COMPARISON

MFA Method	Factor Used	Advantages	Limitations
SMS OTP	Something you have	Easy and widely available	Vulnerable to phishing and SIM-related attacks
Authenticator App	Something you have	Convenient and generally stronger than SMS	Device can be lost; codes can be phished
Push Notification	Something you have	Fast and user-friendly	Can be abused through repeated approval requests
Hardware Security Key	Something you have	Strong and phishing-resistant when properly implemented	Can be lost or damaged
Biometric	Something you are	Convenient and difficult to share	Privacy and device-related concerns

Password + Security Key	Know + Have	Strong protection against many account attacks	Requires physical security key
Password + Biometric	Know + Are	Convenient and strong	Depends on device and biometric implementation

15. HOW MFA IMPROVES ACCOUNT SECURITY

MFA improves account security by adding additional protection beyond a password.

Without MFA:

Username + Password → Access

If the password is stolen, an attacker may be able to access the account.

With MFA:

Username + Password → Additional Factor → Access

If the password is stolen, the attacker may still be unable to access the account because the additional factor is required.

16. BENEFITS OF MFA

1. Protects Against Stolen Passwords

If an attacker obtains a password, the second authentication factor can prevent unauthorized access.

2. Reduces Account Takeover

MFA makes it more difficult for attackers to take control of accounts.

3. Provides an Additional Security Layer

MFA adds another barrier between an attacker and the protected system.

4. Protects Sensitive Information

MFA can help protect email, financial, academic, business, and cloud accounts.

5. Helps Organizations

Organizations can use MFA to protect employees, administrators, applications, and sensitive systems.

6. Reduces the Impact of Credential Theft

A stolen password alone may not be sufficient to authenticate when MFA is properly implemented.

17. LIMITATIONS OF MFA

Although MFA significantly improves security, it does not eliminate every cybersecurity risk.

1. Phishing

Attackers may create fake websites that trick users into providing both passwords and authentication codes.

2. MFA Fatigue

Attackers may repeatedly send push notifications hoping that a user accidentally approves one.

3. Device Loss

A user may lose the phone or security device used for authentication.

4. Account Recovery Risks

Weak account recovery procedures can sometimes be abused to bypass authentication.

5. Social Engineering

Attackers may attempt to convince users or support staff to bypass authentication controls.

Therefore, organizations should combine MFA with security awareness, secure recovery processes, monitoring, and other security controls.

18. IS OTP ALONE MFA?

No, an OTP alone is not necessarily MFA.

An OTP is an authentication mechanism, but MFA requires **two or more independent authentication factors from different categories**.

Example 1: Not MFA

OTP only → Access

Only one authentication factor is being used.

Example 2: MFA

Password + OTP sent to a registered device → Access

Here:

- Password = Something you know
- Registered device/OTP method = Something you have

Therefore, this can constitute MFA when the factors are independently verified.

Important Point

Two authentication methods do not automatically mean two-factor authentication. The methods must represent different factor categories.

19. MFA IMPLEMENTATION IN AN ORGANIZATION

An organization can implement MFA using the following process:

Step 1: Identify Important Accounts

Identify accounts that contain sensitive information or have administrative privileges.

Step 2: Select an Appropriate MFA Method

Choose suitable authentication methods such as authenticator applications, hardware security keys, or biometrics.

Step 3: Enroll Users

Register employees and their approved authentication devices.

Step 4: Configure Authentication Policies

Define when MFA is required and which authentication methods are permitted.

Step 5: Provide Backup Methods

Provide secure recovery procedures for situations where users lose their authentication device.

Step 6: Train Employees

Teach employees how to recognize suspicious authentication requests and phishing attempts.

Step 7: Monitor Authentication Activity

Monitor login attempts and investigate unusual authentication behavior.

20. MFA BEST PRACTICES

Organizations should follow these practices when implementing MFA:

1. Enable MFA for important accounts.
2. Prioritize administrator and privileged accounts.
3. Prefer phishing-resistant authentication methods where practical.
4. Avoid relying solely on SMS when stronger options are available.

5. Educate users about phishing and MFA fatigue.
 6. Secure account recovery processes.
 7. Maintain backup authentication methods.
 8. Monitor suspicious login attempts.
 9. Keep authentication software and devices updated.
 10. Review MFA configurations regularly.
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21. REAL-LIFE EXAMPLE

Scenario

An employee's password is stolen through a phishing attack.

Without MFA

The attacker enters:

Username + Stolen Password

The attacker may gain access to the employee's account.

With MFA

The attacker enters:

Username + Stolen Password

The system then requests an additional authentication factor.

The attacker does not possess the employee's registered authentication device or security key.

Result: Access can be blocked.

This demonstrates how MFA can reduce the risk associated with stolen passwords.